# Network Science

## Lecture 08: Network Dynamics

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# From Structure to Dynamics

Lectures 03-07 analyzed networks as static objects: ties, bridges, communities, balance, short paths, hubs, and small-world shortcuts.

Today we ask a different question:

> **What happens *on* the network?**

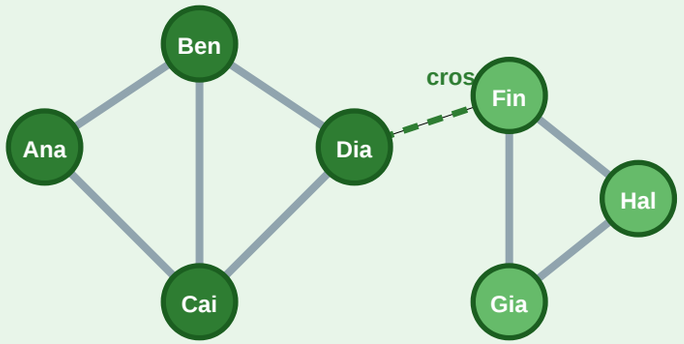
A cold, a rumor, PageRank score, a product failure, and a health behavior can all spread over edges. But they do not follow the same rule.

# Intuition: Same Network, Different Processes

**Workplace contact network: two spreading rules**

### Simple contagion (cold)

one hearing suffices → bridge helps



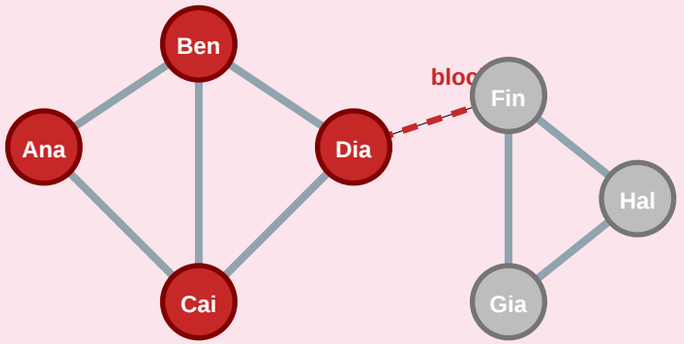

**cold can reach everyone**

Dia exposes Fin; Fin exposes Gia and Hal

weak ties = accelerators

### Complex contagion (behavior)

threshold q = 2 contacts → bridge blocks




**behavior stays in Team A**

Fin needs 2 examples but only sees Dia

weak ties = barriers

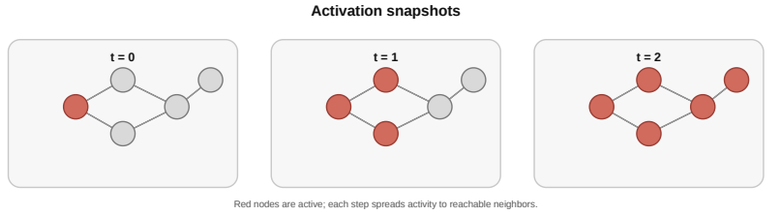
The Dia-Fin bridge helps simple contagion but blocks complex contagion. Structure alone does not determine the outcome - the process matters.

**Cold:** one infectious contact can be enough, so the Dia-Fin bridge can carry the cold to the second team.

**Health behavior:** Fin may need to see two peers join before changing behavior, so the same bridge is too thin. The behavior stalls at the community boundary.

# Four Questions

1. **What spreads?** Signal, probability mass, disease, adoption, failure?
2. **What changes a node?** One exposure, repeated exposure, a threshold, or a flow equation?
3. **Does the state stick?** Forever, temporarily, or not at all?
4. **Which topology helps?** Hubs, bridges, clusters, shortcuts, or communities?



# The Process-Structure Gap

The course has accumulated several gaps between clean theory and real data. L08 adds the **process-structure gap**.

| Lecture | Gap type | Core issue |
|---|---|---|
| L03-L04 | Computational | exact ideals vs. feasible proxies |
| L05 | Causal | snapshots do not identify mechanisms |
| L06 | Structural | complete-graph theory vs. sparse data |
| L07 | Navigational | short paths exist but may not be findable |
| **L08** | **Process-structure** | same graph, different outcomes for different processes |

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| **L08** | **Process-structure** | same graph, different outcomes for different processes |

Network structure is necessary but not sufficient. A spreading model needs both a graph and a rule.

# Takeaway

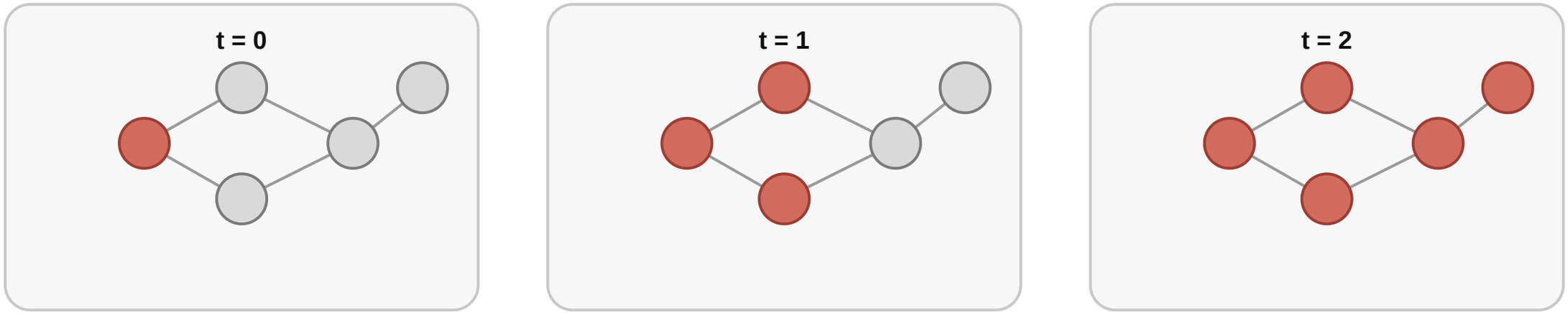
> Network dynamics ask how states, signals, behaviors, or risks move over edges. The right model depends on the spreading rule: flow, cascade, threshold, infection, recovery, persistence, and timing.

# Basic Network Diffusion Models

**Guiding Question:** How can we model the spread of a signal, activation, or influence before we specialize to epidemics?

# Intuitive Example: A Message Starts at A

**Activation snapshots**




Red nodes are active; each step spreads activity to reachable neighbors.

> A message starts at one seed node. It may move like a wave, leak like probability mass, trigger each neighbor independently, or require social reinforcement.

The graph gives possible channels. The diffusion rule says which channels are actually used.

# Case 1: Deterministic Reachability

Start with the simplest possible rule:

> If a node is active, it activates **all** of its inactive neighbors in the next step.

This is the BFS idea from L02 turned into a spreading process.

| Time | Active nodes |
|---|---|
| $t = 0$ | $A$ |
| $t = 1$ | $A, B, C$ |
| $t = 2$ | $A, B, C, D$ |
| $t = 3$ | $A, B, C, D, E$ |

# Why Reachability Is Too Strong

In real spreading, two things often fail:

1. **Not every neighbor is contacted.** Ana may tell only one person, not everyone.
2. **Not every contacted neighbor responds.** A person may ignore a message or not click a link.

> So we need a softer model: instead of activating all neighbors, let the spread choose paths probabilistically.

**Application.** Browsing, attention, and search behavior: users follow one link at a time, not every outgoing link at once.

# Case 2: Random-Walk Diffusion

A random walker sits on one node. At each step, it chooses one outgoing edge and moves along it.

If node $j$ has out-degree $d_j$, then each outgoing neighbor receives probability $1/d_j$.

For the transition matrix $P$:

- $P_{ij} = 1/d_j$ if there is an edge $j \to i$
- $P_{ij} = 0$ otherwise

$P$ is a **transition matrix**: column $j$ says where probability moves next from node $j$.

For $A \to B$, $A \to C$, $B \to D$, $C \to D$:

| $P$ | from A | from B | from C | from D |
|---|---|---|---|---|
| **to A** | 0 | 0 | 0 | 0 |
| **to B** | 1/2 | 0 | 0 | 0 |
| **to C** | 1/2 | 0 | 0 | 0 |
| **to D** | 0 | 1 | 1 | 0 |

From $A$, half the mass goes to $B$ and half to $C$.

# Random-Walk Theory: Reading the State Vector

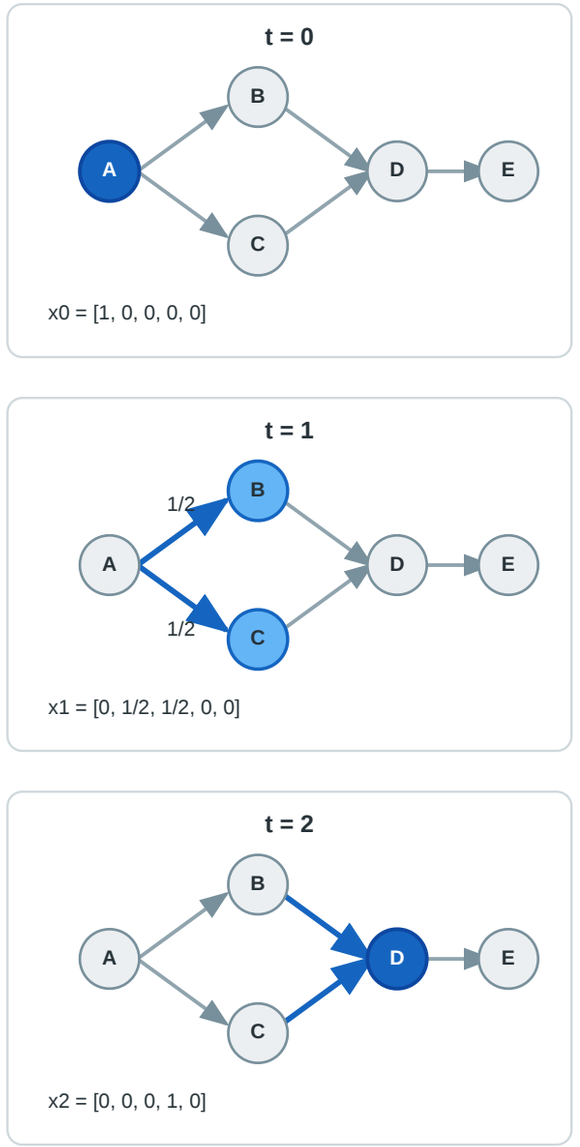
# The probability state vector $x_t$ has one entry per node.

If the node order is $(A, B, C, D, E)$, then

$$x_t = [0, 1/2, 1/2, 0, 0]$$

means: at time $t$, the walker is at $B$ with probability $1/2$ and at $C$ with probability $1/2$.

The entries sum to $1$ when probability is conserved.



# Random-Walk Theory and Application

If $x_t$ stores the probability mass at time $t$, then:

$$x_{t+1} = Px_t.$$

> The process no longer asks "which nodes become active?" It asks "where is the probability mass after repeated movement?"

**Example.** Starting from $A$, probability splits between $B$ and $C$, then recombines at $D$.

**Application.** Web browsing, PageRank-like importance, traffic flow, spreading activation in search and recommendation.

# Why Plain Random Walks Need Damping

A plain random walk can drift too far from the original topic or get trapped in a sink.

> Imagine searching for "network diffusion." Following links forever may eventually land on a barely related page. A useful model should sometimes return to the original query or seed node.

This return is called **damping**, **restart**, or **teleportation**, depending on the application.

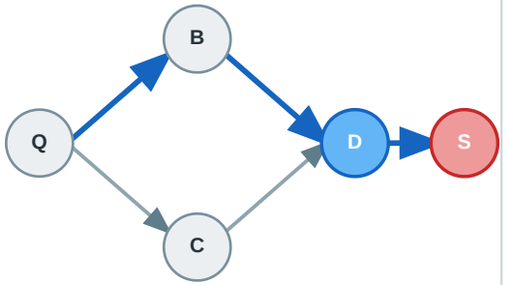
**Application.** Personalized search: explore nearby pages, but keep pulling attention back to the user's original interest.

**Link following with and without restart**

Q is the original query/seed.

**Plain walk**

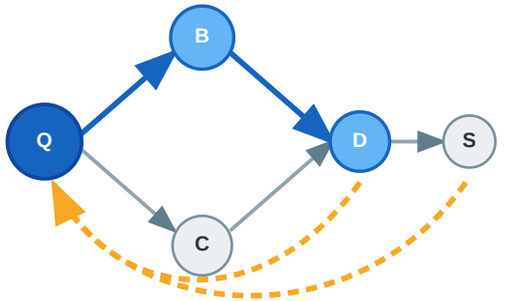
x(t+1) = P x(t)




Mass can drift into a sink/trap.

**Damped walk**

alpha follows links; 1-alpha restarts




Restart keeps mass near x0.

# Case 3: Damped Diffusion

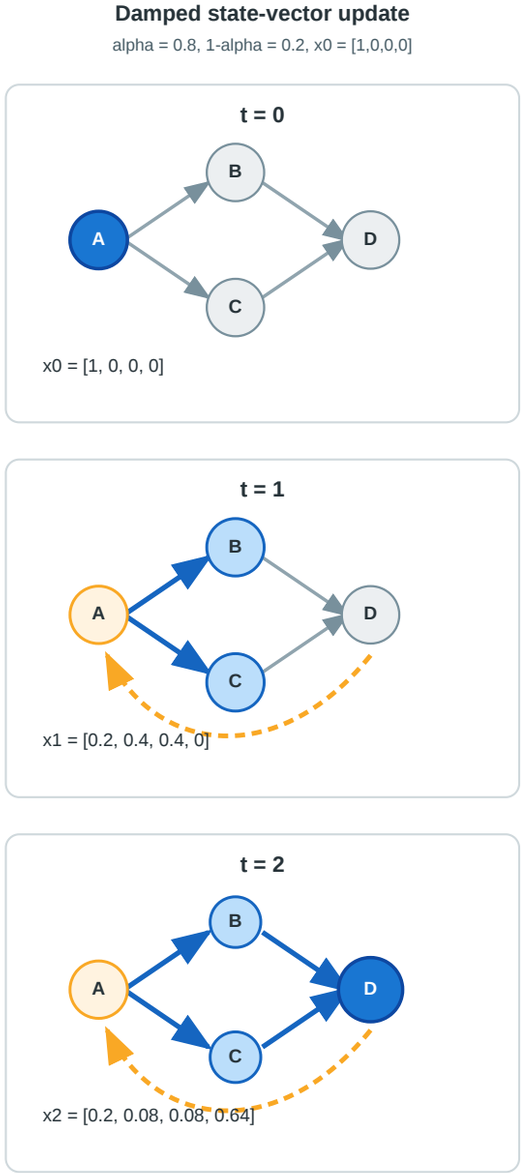
A damped diffusion mixes two actions:

1. **follow the network** with probability $\alpha$
2. **restart** at the seed distribution $x_0$ with probability $1 - \alpha$

$$x_{t+1} = \alpha P x_t + (1 - \alpha) x_0, \qquad 0 < \alpha < 1.$$

$x_0$ is the **restart distribution**: it says where the walk returns when it restarts. For personalized search, $x_0$ is concentrated on the query or seed nodes.

$\alpha$ is the **link-following probability**. Large $\alpha$: explore farther. Small $\alpha$: stay close to $x_0$.




PageRank uses this random-surfer idea for ranking Web pages (Brin & Page, 1998).

# Why Random Walks Are Still Too Smooth

Random walks describe attention or probability mass. But many social processes are **events**:

- a person shares a post or does not
- a customer buys or does not
- a device fails or does not

> For event-like spread, we need a model where each exposure can succeed or fail.

That gives the **independent cascade** model.

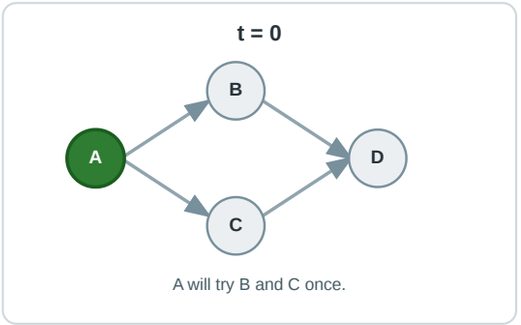
# Case 4: Independent Cascade

Each newly active node gets **one chance** to activate each inactive neighbor.
For an edge $u \to v$, activation succeeds with probability $p_{uv}$.

Independent cascade is **not** a random walk. The active node does **not** choose one outgoing edge. It tries **all** outgoing edges once, and each edge succeeds or fails independently. Therefore one active node can activate many neighbors in the same step, or none.

**Example.** Ana posts a link. Ben clicks and shares with probability $0.3$; Cai clicks and shares with probability $0.1$. If Ben does not share now, Ana does not keep retrying forever.
**Application.** Viral marketing, information cascades, link sharing, one-shot recommendations. Kempe, Kleinberg & Tardos (2003) use this model for influence maximization.

**One chance per outgoing edge**
each edge succeeds or fails independently



# Why One Exposure Is Sometimes Not Enough

Some behaviors are not triggered by one contact:

- joining a protest may require seeing several friends join
- registering for a health program may require social proof
- changing a habit may require repeated reinforcement

> Here the important signal is not "did one edge transmit?" but "how much of my neighborhood is already active?"

That gives the **linear threshold** model.

# Case 5: Linear Threshold

Node $v$ activates when enough weighted active neighbors point to it:

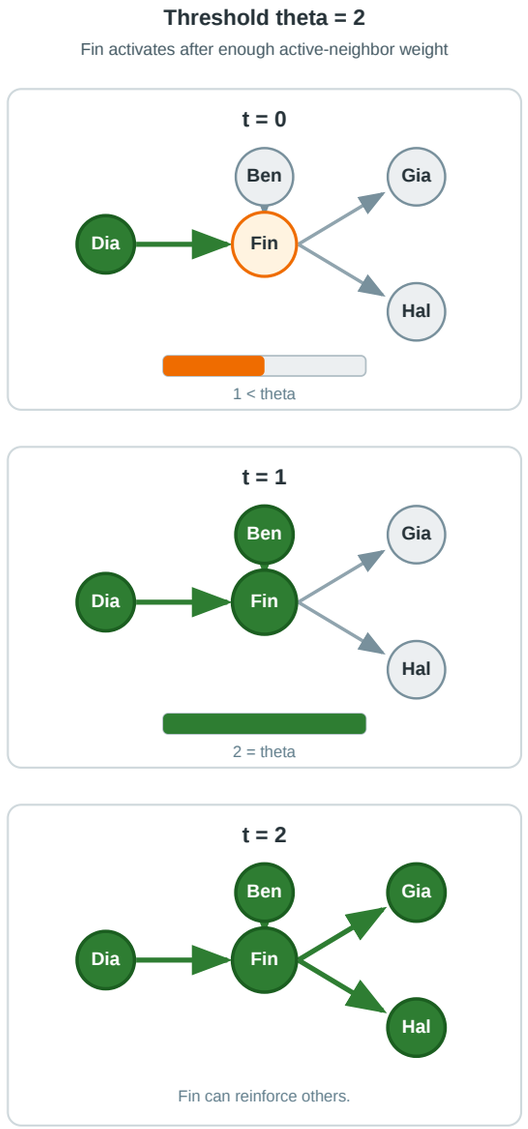
$$\sum_{u \in N(v)} w_{uv} x_u \geq \theta_v.$$

$x_u = 1$ means neighbor $u$ is active; $\theta_v$ is $v$'s adoption threshold.

**Example.** Fin joins a health program only after seeing at least two close colleagues join.

**Application.** Behavior change, norm adoption, political mobilization, technologies that need coordination.

Granovetter (1978) formalized threshold models of collective behavior; Kempe, Kleinberg & Tardos (2003) use linear threshold models for influence spread.



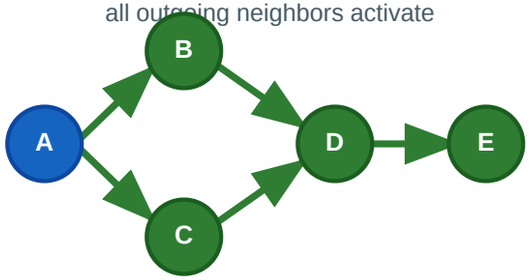
# Practical Comparison: Same Graph, Different Rules

## Same graph, five different diffusion rules

A starts the process. The graph is fixed; only the spreading rule changes.

### 1. Reachability

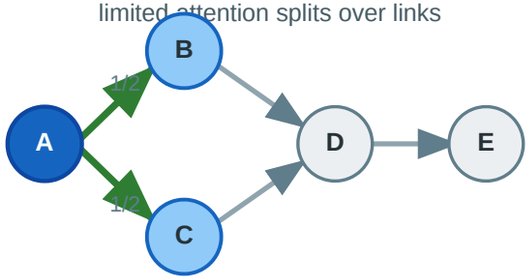
all outgoing neighbors activate




If a path exists, the signal eventually follows it.

### 2. Random walk

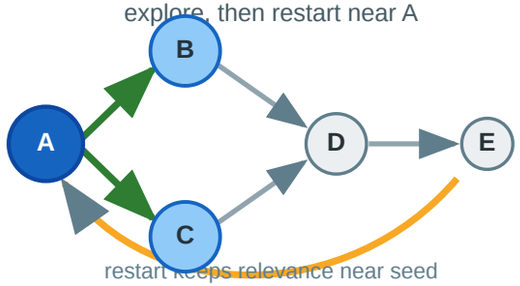
limited attention splits over links




Mass moves probabilistically, not all at once.

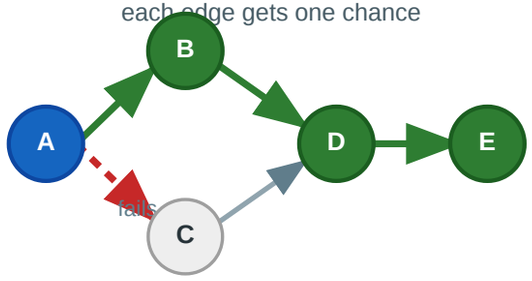
### 3. Damped diffusion

explore, then restart near A




### 4. Independent cascade

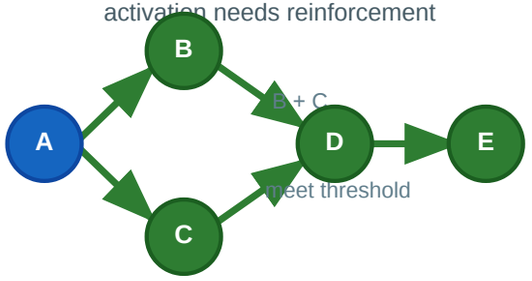
each edge gets one chance




Spread is binary and stochastic.

### 5. Linear threshold

activation needs reinforcement




One exposure may be insufficient.

The graph is fixed. Reachability activates every reachable node; random walks move probability mass; damping keeps diffusion close to the seed; independent cascade makes each edge uncertain; threshold

# Applications: Which Diffusion Rule Fits?

| Application | Natural model | Why |
|---|---|---|
| Web ranking | damped random-walk diffusion | attention flows through links with teleportation |
| Viral post | independent cascade | one exposure can trigger sharing |
| Health behavior | threshold model | peers provide social proof |
| Search over related concepts | diffusion with decay | nearby concepts should count more |
| Failure propagation | cascade model | one failed dependency can trigger another |

The modeling choice is substantive. Choosing a rule means making a claim about how exposure changes behavior.

# Module Takeaway

Basic diffusion models differ in what a node state means: binary activation, probability mass, graded attention, one-shot influence, or threshold behavior. Epidemic models are one important special case, not the whole category.

# Epidemic Models and State Memory

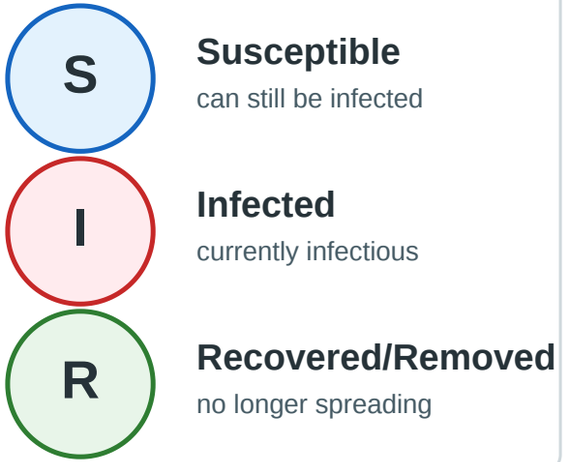
**Guiding Question:** What changes when diffusion is modeled as infection, recovery, immunity, or reinfection?

# Epidemic State Vocabulary

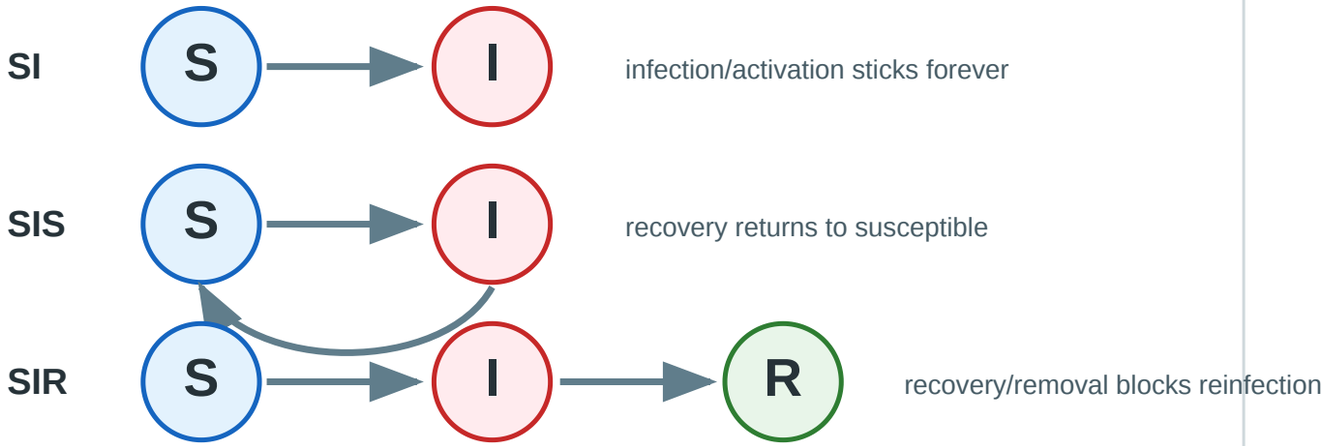
## States and allowed transitions

The model abbreviation tells us which state transitions are allowed.

### States




### Allowed transitions




Bridge from basic diffusion: inactive -> active is SI written in epidemic notation.

The acronym names the allowed path through states: **SI** means $S \to I$, **SIS** means $S \to I \to S$, and **SIR** means $S \to I \to R$.

# Case 1: SI

> **SI:** $S \to I$
>
> Susceptible nodes become infected. There is **no recovery** and no removal.

**Intuition.** Infection or activation accumulates. Once a node enters $I$, it stays in $I$ and can keep transmitting.

**Example.** On a path, an infection starting at $B$ grows outward until all reachable nodes are infected.

**Application.** Cumulative awareness, irreversible adoption, permanent compromise, reachability with epidemic labels.

# SI: Formal Network Update

Let $G = (V, E)$ and let $X_i(t) \in \{S, I\}$ be the state of node $i$ at time $t$.

For a susceptible node $i$, define its infected neighbors:

$$N_I(i, t) = \{j \in N(i) : X_j(t) = I\}.$$

With edge transmission probability $\beta$, the discrete-time update is:

$$\Pr[X_i(t+1) = I \mid X_i(t) = S] = 1 - (1 - \beta)^{|N_I(i,t)|}.$$

The equation says: node $i$ becomes infected unless all currently infected neighbors fail to transmit.

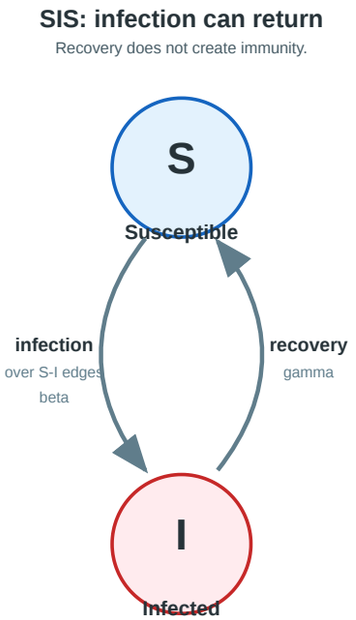
Algorithmically, this is a **stochastic simulation**: at each step, every infected node tries to infect susceptible neighbors. Infected nodes never leave $I$.

This is the epidemic version of **Case 1: Deterministic Reachability** if activation is certain; with probabilistic one-shot edge attempts, it becomes close to **Case 4: Independent Cascade**.

# Case 2: SIS

**SIS:** $S \to I \to S$

Infected nodes recover, but recovery returns them to the susceptible state.




**Intuition.** The infection can come back. A node that recovered is not protected.

**Example.** In a small group, Ana recovers from a cold but later catches it again from Ben.

**Application.** Recurring infections without lasting immunity, malware that can be removed and reinstalled, repeated misinformation exposure.

# SIS: Formal Network Update

SIS uses the same infection step as SI, but infected nodes can recover:

$$\Pr[X_i(t+1) = I \mid X_i(t) = S] = 1 - (1-\beta)^{|N_I(i,t)|}$$

$$\Pr[X_i(t+1) = S \mid X_i(t) = I] = \gamma.$$

$\gamma$ is the recovery probability per time step: an infected node returns to $S$ with probability $\gamma$ and remains infected with probability $1 - \gamma$.

Algorithmically: infection and recovery are repeated forever. A simulation stops only when the infection dies out, reaches a time limit, or approaches a stable endemic level.

**Endemic** means the infection does not disappear, but remains present at a roughly stable nonzero level.

# Case 3: SIR

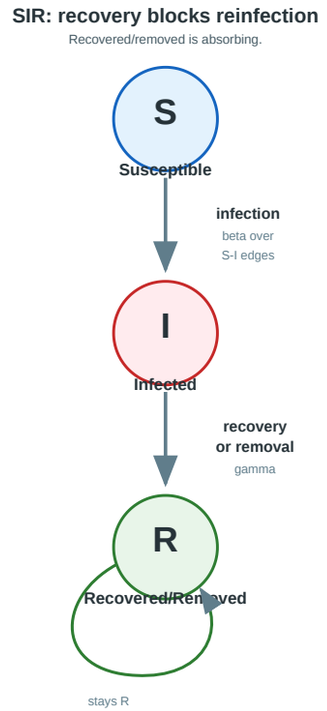
**SIR:** $S \to I \to R$

Infected nodes recover into a recovered/removed state. They no longer transmit and cannot be infected again in the model.

**Intuition.** The infection burns through reachable parts of the graph and leaves immune or removed nodes behind.

**Example.** A one-time infection moves along a path as a wave: current cases are $I$, past cases are $R$.

**Application.** Diseases with immunity, one-shot outbreaks, finite attention cascades, final outbreak size.



# SIR: Formal Network Update

SIR uses infection plus one-way removal:

$$\Pr[X_i(t+1) = I \mid X_i(t) = S] = 1 - (1-\beta)^{|N_I(i,t)|}$$

$$\Pr[X_i(t+1) = R \mid X_i(t) = I] = \gamma.$$

Recovered or removed nodes stay removed:

$$\Pr[X_i(t+1) = R \mid X_i(t) = R] = 1.$$

# SIR: What We Compute

SIR is used for time-dependent outbreak quantities:

| Quantity | Meaning |
|---|---|
| $I(t)$ | current number or fraction infected |
| $\max_t I(t)$ | peak load |
| time of peak | when capacity pressure is highest |
| $R(\infty)$ | final outbreak size |
| extinction probability | probability the outbreak dies early |

> On networks, exact closed-form solutions are rare. We usually simulate the stochastic process, approximate it with aggregate equations, or use percolation-style results for final size (Kermack & McKendrick, 1927; Newman, 2002; Van Mieghem, 2014).

# Percolation View of SIR Final Size

Percolation removes the time ordering and asks a simpler reachability question:

> Keep each edge with probability $T$, where $T$ is the chance that infection would successfully cross that edge before recovery.

How to use it:

- Sample each edge independently: open with probability $T$, closed with probability $1 - T$.
- In this sampled graph, compute $R(\infty)$ as the nodes reachable from the seed through open edges.
- Repeat both steps many times to estimate expected final size and variability.

$R(\infty)$ is not automatically all nodes; it is the reachable open-edge component from the seed in one sampled graph.

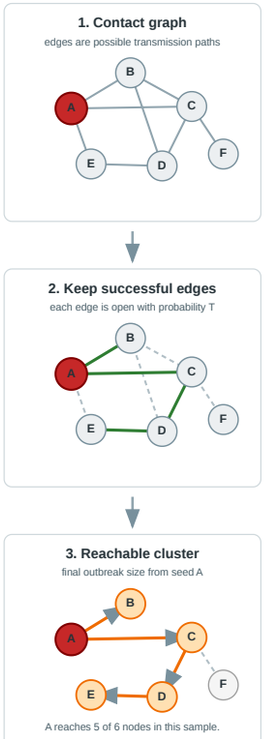
Common ways to compute $T$:

- **continuous-time rates:** infection and recovery are competing clocks:

$$T = \frac{\beta}{\beta + \gamma}.$$

- **discrete fixed duration:** time is counted in steps, for example days:

$$T = 1 - (1 - p)^{\tau}.$$



# What Percolation Measures

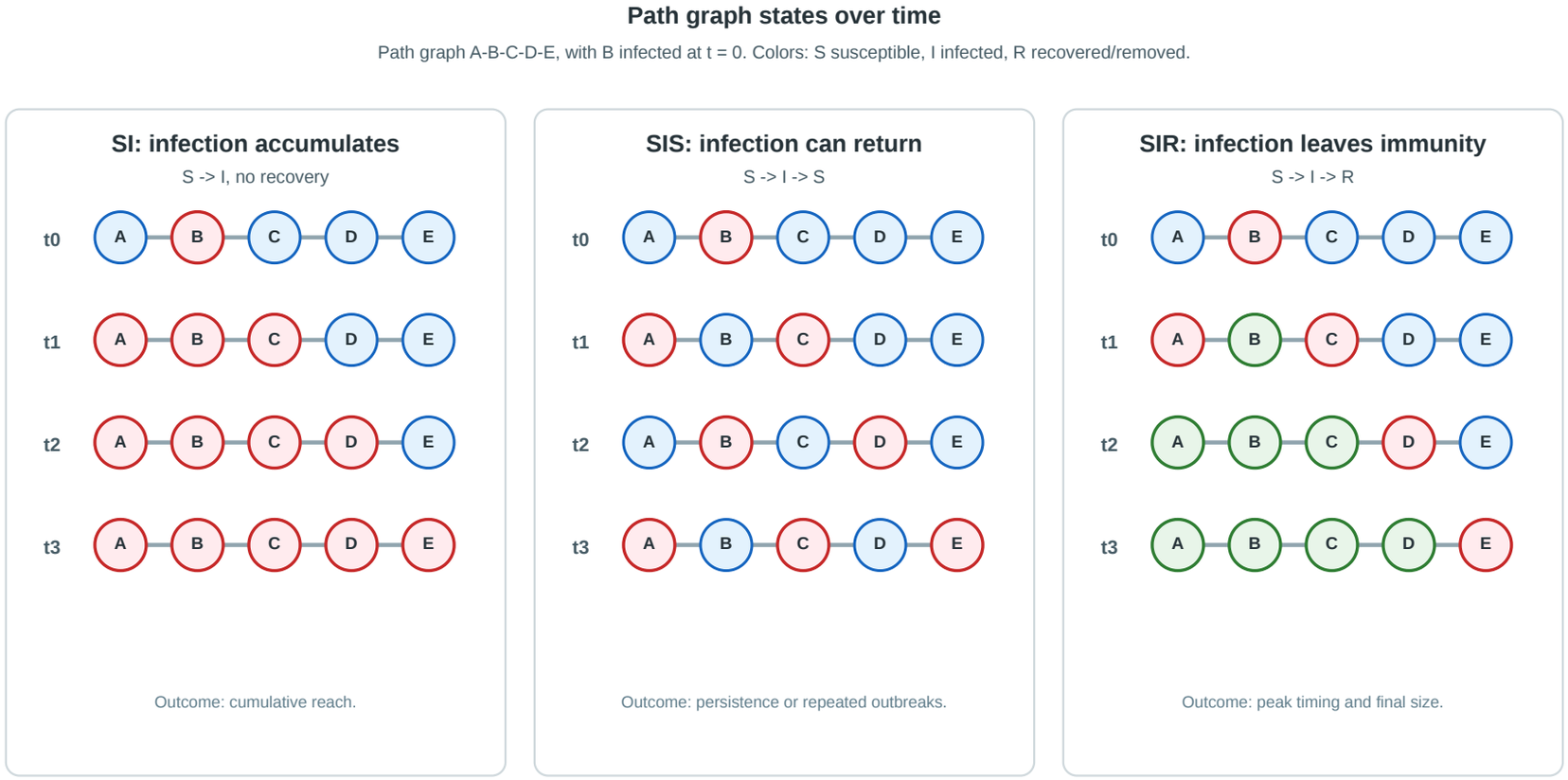
Percolation is useful when we care about **final reach**, not the full time curve.

| Object | Meaning in SIR |
|---|---|
| open edge | infection would cross this contact before recovery |
| closed edge | infection would fail to cross this contact |
| reachable component | nodes eventually infected from the seed |
| giant component | outbreak can reach a macroscopic part of the network |
| threshold $T_c$ | graph-level critical transmissibility where large outbreaks become possible |

Seed choice depends on the question: use an observed first case, sample random introductions, or compare high-risk seeds such as hubs and bridge nodes.

> Percolation is not a node-state simulation. It is a structural shortcut for estimating final outbreak size under an SIR-like one-shot transmission process.

# Practical Example: Same Path, Three Epidemics




The graph and seed are the same. The only difference is the memory rule after infection: stay infected, return to susceptible, or move to recovered/removed.

# Aggregate View: From Nodes to Fractions

The network model stores a state for every node:

$$X(t) = (X_1(t), X_2(t), \ldots, X_n(t)).$$

The network simulation tracks every node state. The **aggregate view** compresses those node states into population fractions:

$$S(t) = \frac{|\{i : X_i(t) = S\}|}{n}, \quad I(t) = \frac{|\{i : X_i(t) = I\}|}{n}, \quad R(t) = \frac{|\{i : X_i(t) = R\}|}{n}.$$

So $S(t)$, $I(t)$, and $R(t)$ are proportions between 0 and 1, and $S(t) + I(t) + R(t) = 1$. Raw counts are still possible, but the SIR/SIS approximation slides use fractions.

Aggregation answers "how many?" It no longer answers "which node?", "which bridge?", or "which community?"

# Aggregate View: SIS Approximation

A **well-mixed** approximation ignores the graph and assumes every infected node can contact every susceptible node with equal chance.

For SIS, write $I(t)$ for the infected **fraction** and $S(t) = 1 - I(t)$.

$$\frac{dI}{dt} = \beta SI - \gamma I = \beta I(1 - I) - \gamma I.$$

This is an **ordinary differential equation (ODE)**: it describes how one quantity changes continuously over time. It is not the closed-form solution; the right-hand side gives the instantaneous rate of change of the infected fraction.

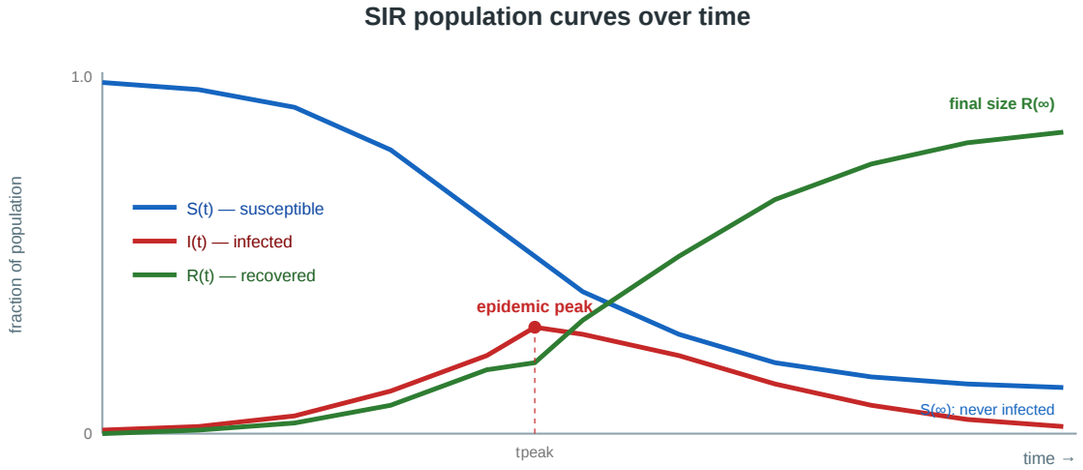
Infection grows through $\beta SI$ and shrinks through recovery $\gamma I$. In the simplest well-mixed form, persistence requires roughly $\beta/\gamma > 1$.

# Aggregate View: The SIR Curve

In the well-mixed SIR approximation, $S(t)$, $I(t)$, and $R(t)$ are **fractions of the population**:

$$\frac{dS}{dt} = -\beta SI$$
$$\frac{dI}{dt} = \beta SI - \gamma I$$
$$\frac{dR}{dt} = \gamma I.$$

$\beta$ is the infection rate, $\gamma$ is the recovery rate; assumes homogeneous mixing, not explicit edges.




Read the chart as three aggregate time series: $S(t)$ falls as susceptible people are infected, $I(t)$ rises while new infections exceed recoveries and then falls after the peak, and $R(t)$ accumulates everyone who has recovered or been removed. $R(\infty)$ is the final outbreak size; $S(\infty)$ is the fraction that escaped infection.

The basic reproduction number is: $R_0 = \beta/\gamma$.

It is the expected number of secondary infections caused by one infected case in a fully susceptible well-mixed population. If $R_0 > 1$, early growth is possible; if $R_0 < 1$, infections tend to die out.

# SIR: Closed Form or Approximation?

The standard well-mixed SIR equations usually do **not** give a simple closed-form expression for $S(t)$, $I(t)$, and $R(t)$ over time.

> In practice, we usually compute SIR curves by numerical integration of the ODEs, meaning ordinary differential equations, or by stochastic simulation on a graph.

> This does not mean "theory is useless." It means analytical results usually answer specific questions rather than giving the full curve in one formula.

# Analytical Results: Threshold and Peak

## Threshold intuition

At the beginning, almost everyone is susceptible. The central question is:

> Does one infected case create more than one new infected case on average?

In the simplest well-mixed SIR model, this is summarized by $R_0 = \beta/\gamma$.

If $R_0 > 1$, early growth is possible. If $R_0 < 1$, the outbreak tends to die out.

## Peak condition

The infected curve peaks when it stops increasing:

$$\frac{dI}{dt} = 0.$$

For well-mixed SIR:

$$\frac{dI}{dt} = \beta SI - \gamma I.$$

So the peak occurs when infection pressure and recovery balance:

$$\beta S = \gamma.$$

> Example: if $\beta = 0.30$ and $\gamma = 0.10$, then $\gamma/\beta = 1/3$. Infections rise while $S(t) > 1/3$ and peak when the susceptible fraction has fallen to about one third.

# Analytical Results: Final Size and Uncertainty

## Final-size relation

Even without a closed form for the whole curve, SIR can characterize the final outbreak size implicitly. Implicit means the result is defined by an equation that usually still needs numerical solving.

## Definition

**Final outbreak size** is defined as:

$$R(\infty) = \frac{|\{i \in V : X_i(\infty) = R\}|}{|V|} = \frac{|\{i \in V : i \text{ was ever infected}\}|}{|V|}.$$

In words: count the nodes that end in state $R$ and divide by the number of nodes.

Here **removed** means "out of the spreading process": no longer infectious and not susceptible again in this SIR model.

Usually $R(\infty) < 1$ because some susceptible nodes are never reached:

$$S(\infty) + R(\infty) = 1.$$

So final size is not current infection. It is cumulative reach.

# Simulation or Closed Form?

There are three common computational levels:

| Level | Representation | What it gives |
|---|---|---|
| stochastic network simulation | state of every node | sample paths, uncertainty, seed effects |
| aggregate ODE (ordinary differential equation) | counts/fractions $S(t), I(t), R(t)$ | smooth curves, thresholds, peak intuition |
| network approximation | degree moments, eigenvalues, percolation | topology-sensitive thresholds/final size |

> Closed-form solutions are limited. For real networks, simulation is the default; formulas are used to understand regimes and check intuition.

# Network Simulation Template

For SI, SIS, and SIR, the simulation pattern is nearly the same:

```
initialize node states
for t = 1 ... T:
  collect infection attempts over S-I edges
  apply successful infections with probability beta
  apply recovery/removal with probability gamma
  record S(t), I(t), R(t)
```

> The model-specific part is the post-infection state: SI keeps nodes infected, SIS returns them to susceptible, SIR moves them to recovered/removed.

# Simulation Stability: What To Expect

Stochastic epidemic simulations are not single deterministic answers. Repeated runs can differ, especially near the epidemic threshold.

| Model | Expected long-run behavior |
|---|---|
| SI | monotone growth until no reachable susceptible nodes remain |
| SIS | extinction or noisy persistence around an endemic level |
| SIR | one outbreak wave, then absorption when $I(t) = 0$ |

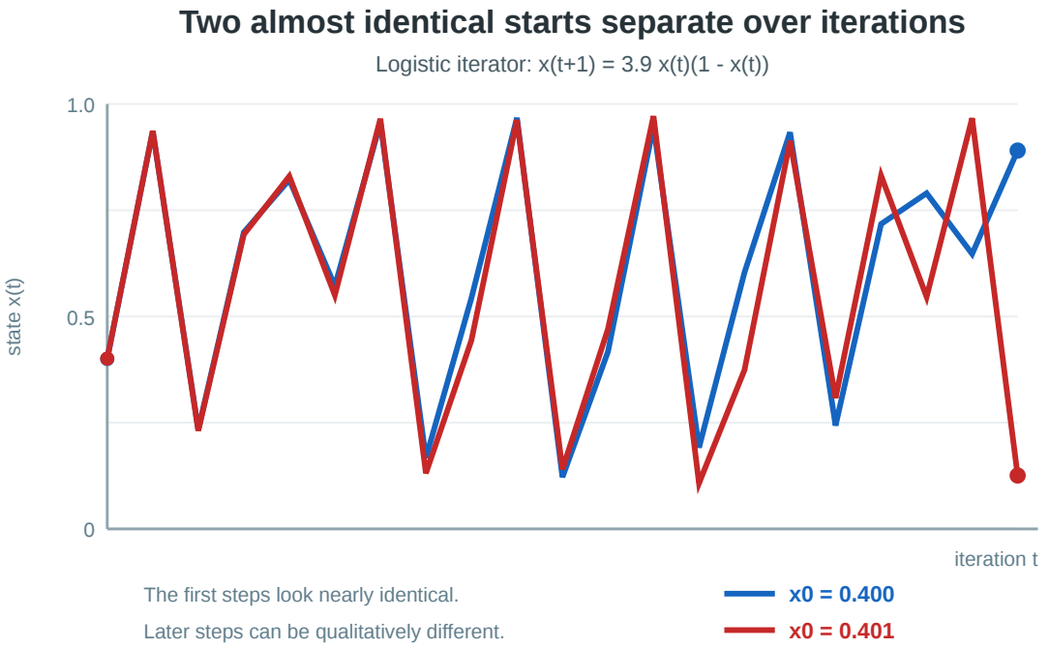
> Near a threshold, small random differences can decide whether the infection dies early or becomes large. Report several runs, averages, and uncertainty bands rather than one trajectory.

# Numerical Stability: Iteration Can Amplify Error

Even a simple iterative rule can become numerically sensitive:

$$x_{t+1} = r x_t (1 - x_t).$$

For some values, such as $r = 3.9$, two almost identical starting values can separate quickly (May, 1976).




> This logistic iterator is not an epidemic model. It is a warning: simulations are repeated numerical updates, and repeated updates can amplify rounding, step-size, and modeling choices.

# Reproducibility: Seeds, Precision, Hardware

Reproducibility also depends on random seeds. Fellicious, Weissgerber & Granitzer (2020) show this for CNN accuracy: reporting a single run can hide relevant variation caused by initialization and training randomness.

On GPUs and parallel hardware, precision and operation order add another layer: floating-point reductions can produce slightly different results even when the mathematical formula is the same (Goldberg, 1991; Higham, 2002).

For simulation studies, this means:

- the random seed controls one possible trajectory through the stochastic process;
- the time step controls how the continuous process is discretized;
- CPU/GPU and precision settings can change accumulated rounding behavior;
- parallel update order can change results when operations are not exactly associative.

A robust claim should survive reasonable changes in seed, step size, precision, and hardware.

# Numerical Stability: Practical Checks

For epidemic simulations, check robustness before interpreting a single curve:

| Check | Why it matters |
|---|---|
| repeat many runs | separates typical behavior from one random path |
| report uncertainty bands | shows threshold sensitivity and stochastic spread |
| vary time step or update rule | catches discretization artifacts |
| keep probabilities in $[0, 1]$ | prevents invalid numerical states |
| compare CPU/GPU or precision settings | detects hardware and floating-point effects |
| record random seeds | makes results reproducible |
| compare to aggregate intuition | catches obvious implementation errors |

Numerical stability is not just "no crash." It means the qualitative conclusion survives reasonable changes in resolution, seed, and implementation choices.

# Module Takeaway

> Epidemic models are diffusion models with explicit state transitions. The crucial modeling decision is state memory: infected forever (SI), recover and become susceptible again (SIS), or recover into removal/immunity (SIR).

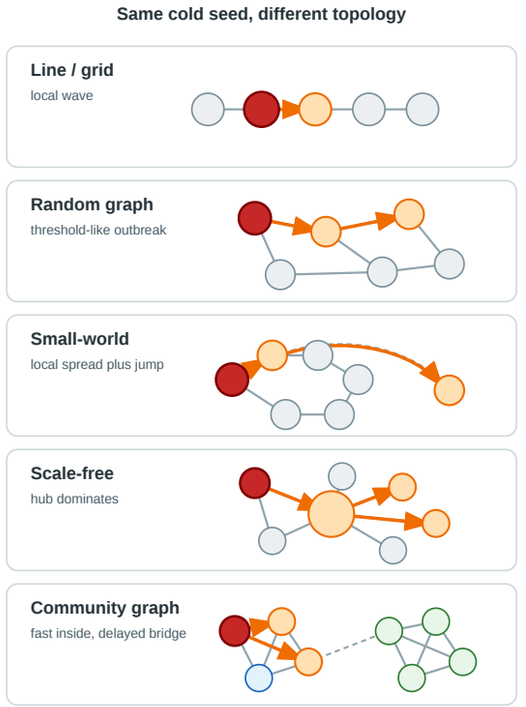
# Effects of Network Topology

**Guiding Question:** How does the same spreading rule behave on different graph families?

# Intuitive Example: A Cold Starts at One Node

Imagine the same cold with the same per-edge transmission tendency and recovery behavior starts in five different networks:

| Topology | First intuition |
|---|---|
| line or grid | slow wave |
| random graph | threshold-like outbreak |
| small-world | local spread plus long jumps |
| scale-free | hubs dominate |
| community graph | fast inside, delayed between groups |



# Homogeneous Networks: SIR

In relatively homogeneous networks, most nodes have similar degree. Average degree is therefore a useful first approximation: unlike a well-mixed model, contacts are still edges, but $\langle k \rangle$ summarizes the typical number of neighbors that one infectious node can try to infect.

From here on, use edge-level parameters: $\beta_e$ is the infection rate **per edge**, and $\gamma_e$ is the recovery rate for an infected node.

For **SIR-like outbreaks**, average degree multiplies infection opportunities:

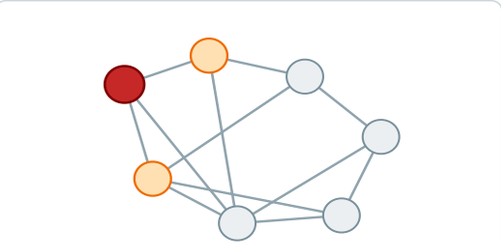
$$R_0 \approx \frac{\beta_e}{\gamma_e}\langle k \rangle \qquad \langle k \rangle = \frac{1}{n}\sum_{i=1}^{n} k_i,$$

Read this as an early-outbreak approximation:

- $\beta_e$ controls how often infection succeeds along one contact edge.
- $\gamma_e$ is the recovery rate; with continuous-time exponential recovery, the expected infectious time is $1/\gamma_e$.

If this product is larger than one, one infected node tends to create more than one new infected node before recovery.

**Homogeneous graph: average degree is informative**



No extreme hub: most nodes have comparable contact opportunities.

For SIR, the main use is outbreak risk and approximate final size: below the threshold most introductions die out; above it, a large recovered/removed set can become possible.

# Homogeneous Networks: SIS (1/2)

SIS has a different question than SIR. Because recovered nodes become susceptible again, we do not mainly ask "how large was the final outbreak?" We ask whether infection can persist as a long-running endemic process.

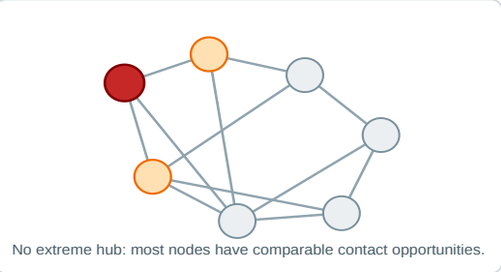
For a homogeneous graph, the network threshold approximation becomes:

$$\frac{\beta_e}{\gamma_e} \gtrsim \frac{1}{\langle k \rangle}.$$

Persistence becomes plausible when per-edge infection strength, multiplied by the average number of contact opportunities, exceeds recovery.

If $\langle k \rangle$ is larger, each infected node has more chances to reintroduce infection before recovery. Therefore the infection rate needed for persistence is lower.

**Homogeneous graph: average degree is informative**



No extreme hub: most nodes have comparable contact opportunities.

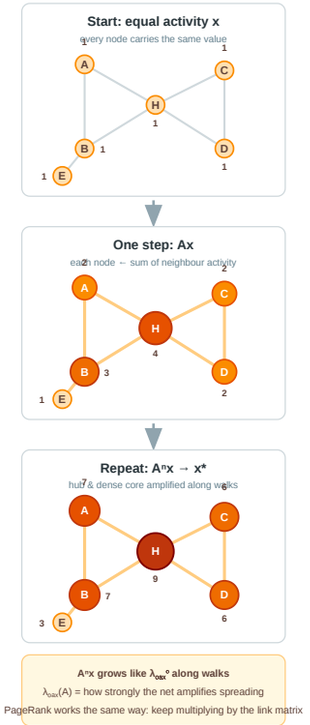
# Homogeneous Networks: SIS (2/2)

For a general network, average degree can be too crude. A common network threshold intuition is:

$$\frac{\beta_e}{\gamma_e} \gtrsim \frac{1}{\lambda_{\max}(A)}.$$

Here $\lambda_{\max}(A)$ is the largest eigenvalue of the adjacency matrix. For its leading eigenvector $x$:

$$Ax = \lambda_{\max} x.$$




**Why?** $\lambda_{\max}$ is the largest factor by which multiplying a nonzero activity vector by $A$ can stretch it. Since $Ax$ replaces each node's activity by the sum of activity coming from its neighbors, repeated multiplication models repeated spreading along walks. If the graph contains hubs, dense cores, or many mutually reinforcing paths, this neighbor-summing grows strongly. That is why $\lambda_{\max}(A)$ measures how much the network can amplify reinfection. This is exactly the power iteration you already know from **PageRank**, where the same repeated matrix multiplication converges to the leading eigenvector.

An **endemic** SIS process means the infection does not simply burn out; it remains present for a long time at a nonzero infected fraction, even though individual nodes recover and become susceptible again.

# Heterogeneous Networks and Hubs

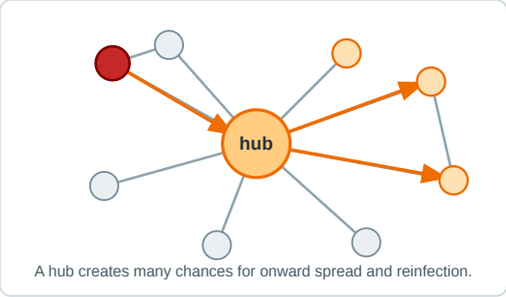
In heterogeneous networks, $\langle k \rangle$ can be misleading because infections are more likely to reach high-degree nodes.

For SIR on uncorrelated random networks, a common threshold approximation uses **transmissibility** $T$: the probability that infection crosses an edge before recovery.

$$T_c \approx \frac{\langle k \rangle}{\langle k^2 \rangle - \langle k \rangle}.$$

For a graph with $n$ nodes and degrees $k_i$: $\langle k^2 \rangle = \frac{1}{n} \sum_{i=1}^{n} k_i^2$.
$T_c$ is a **network-level** critical edge-transmission probability. If the disease transmissibility $T$ is above this threshold, a large outbreak becomes possible. With continuous-time per-edge rates, one common convention is $T = \beta_e/(\beta_e + \gamma_e)$.

**Heterogeneous graph: hubs lower thresholds**




A hub creates many chances for onward spread and reinfection.

This explains why hubs matter for SIR final outbreak risk: high-degree nodes increase $\langle k^2 \rangle$, which lowers the critical transmissibility $T_c$. Lower $T_c$ means that even weaker per-edge transmission can cross the outbreak threshold.

# Vanishing Threshold: What It Means

The previous slide explained why hubs lower thresholds. A **scale-free** network is the limiting hub-heavy case: its degree distribution has a heavy tail, so a few nodes can have extremely large degree.

Pastor-Satorras & Vespignani (2001) showed that idealized infinite scale-free networks with sufficiently heavy-tailed degree distributions can have a **vanishing epidemic threshold**.

**Mathematical idea**

The second moment squares degrees:

$$\langle k^2 \rangle = \frac{1}{n} \sum_i k_i^2.$$

So very large hubs count disproportionately. In an idealized infinite scale-free network with a sufficiently heavy tail, larger and larger hubs keep appearing as $n$ grows, and $\langle k^2 \rangle$ can grow without bound. Then thresholds such as

**Interpretation**

There is no positive critical value that the disease must exceed in the infinite-network limit.

Even weak transmission can be above threshold because hubs create many repeated spreading opportunities.

$$T_c \approx \frac{\langle k \rangle}{\langle k^2 \rangle - \langle k \rangle}$$

can approach zero.

> This does **not** mean parameters such as $\beta_e$, $\gamma_e$, or $R_0$ no longer matter. They still affect speed, prevalence, and finite-network outbreak size. The result says the critical threshold can become extremely small in idealized hub-dominated networks.

# Reading the Hub Result Correctly

**What it says**

- Hubs are infected early with high probability.
- Hubs create many onward transmission opportunities.
- Targeted hub protection can be far more effective than random protection.

**What it does not say**

- Every real disease infects everyone.
- Every social network is scale-free.
- Control is impossible.

> The practical lesson is targeted intervention, not fatalism.

# Theory: Clustering, Shortcuts, and Communities

| Structure | Effect on simple contagion | Effect on threshold / complex contagion |
|---|---|---|
| local clustering | repeated contacts, local saturation | reinforcement, easier adoption |
| weak tie / bridge | fast jump to new group | often too thin to trigger adoption |
| small-world shortcut | collapses path length | helps only if reinforcement follows |
| community boundary | bottleneck for global spread | may trap adoption inside modules |

> Watts & Strogatz (1998) explain how few shortcuts can sharply reduce distances while preserving clustering. Centola (2010) shows that for complex contagion, clustering can beat shorter paths.

- **Simple contagion:** one successful exposure can be enough to transmit, as in a cold or SI/SIR infection.
- **Threshold / complex contagion:** one exposure is usually not enough; adoption needs reinforcement from several neighbors, as in the linear threshold model.

# Practical Example: Intervention by Topology

The same infection model suggests different interventions once the contact graph changes.

| Topology | What to look for | Intervention logic |
|---|---|---|
| grid / spatial contact | local infection front | contain neighborhoods around the boundary before the wave expands |
| random graph | many similar contacts | reduce average transmissibility $T$ or average degree $\langle k \rangle$ |
| scale-free / hub-heavy graph | high-degree nodes and venues | protect hubs first because they lower the epidemic threshold |
| community graph | sparse bridges between groups | monitor or reduce bridge contacts before infection jumps modules |

> The intervention should match the structural bottleneck. A grid is slowed by spatial containment; a hub-heavy graph is slowed by protecting hubs; a modular graph is slowed by controlling bridges.

# Applications: Topology Changes the Question

Topology changes what is worth asking. Before fitting a model, identify which structural feature is likely to dominate the spread.

| Applied setting | Structural feature to inspect | Question topology adds |
|---|---|---|
| public health | households, workplaces, hubs, communities | Will the outbreak stay local, jump communities, or become system-wide? |
| online misinformation | bridges between groups and high-reach accounts | Which accounts or bridges turn local exposure into a cascade? |
| organizational communication | bottlenecks and redundant channels | Does information depend on one bridge, or can it reroute through alternatives? |
| product adoption | clustered communities and wide bridges | Is one exposure enough, or does adoption require local reinforcement? |
| infrastructure failure | dependencies and cascade paths | Which failures disconnect or overload large parts of the system? |



The modeling question becomes: **what spreads, what state memory does it have, and which structural feature controls reach, speed, or persistence?**

# Module Takeaway

> Topology does not merely change the speed of spread. It can change the phase behavior: whether a process dies out, explodes, persists, stalls at a community boundary, or remains local because long-range paths are missing.

# Examples and Applications

**Guiding Question:** How do we choose a diffusion model for a real case?

# Intuitive Example: Model Choice Is a Claim

When we choose a diffusion model, we are not just choosing a formula.

> We are claiming what kind of exposure changes a node.

| If you believe... | Use... |
|---|---|
| one exposure can trigger spread | simple contagion / independent cascade |
| repeated social proof is necessary | threshold / complex contagion |
| infection can return | SIS |
| infection produces immunity/removal | SIR |
| attention flows continuously | random-walk or damped diffusion |

# A Compact Model Map

| Model | Node state | Transition rule | Typical output |
|---|---|---|---|
| Damped diffusion | continuous score | matrix update with restart | ranking / relevance |
| Independent cascade | inactive / active | one probabilistic chance per edge | final adoption set |
| Linear threshold | inactive / active | weighted exposure exceeds threshold | cascade size |
| SI | susceptible / infected | infection without recovery | cumulative reach |
| SIS | susceptible / infected | recovery allows reinfection | persistence |
| SIR | susceptible / infected / removed | infection then removal | peak and final size |

# Empirical Anchor: Centola's Experiment

**Centola (2010),** *The Spread of Behavior in an Online Social Network Experiment*, **Science 329, 1194-1197**.

Centola built an **artificial online health community** and randomly assigned 1,528 participants to anonymous "health buddies." The spreading "behavior" was **voluntarily signing up to use a new online health resource** — an adoption that takes effort, so people tend to join only after **several** neighbors have, not after just one.

# What Centola Found

Centola found that adoption spread farther and faster in clustered networks than in random networks.

> For behavior spreading, clustering beat shortcuts.

Why this matters:

- random networks had shorter paths, which should help simple contagion
- clustered networks supplied repeated social exposure
- adoption was **dose-dependent**: each additional adopting neighbor raised the chance a participant adopted — the signature of reinforcement
- the result supports the complex-contagion interpretation for this behavior

# What the Study Could Not See

**Controlled networks are not natural networks.** Real social networks are heterogeneous, layered, temporal, and adaptive.

**The behavior was binary.** Registration was observed, not intensity or long-term engagement.

**The process had no reversal.** Once registered, participants stayed registered in the measured outcome.

**The result is about one behavior.** Other behaviors may be simpler, more private, cheaper, or more reversible.

# Applications: Pick the Model Before the Metric

| Situation | First modeling question |
|---|---|
| vaccination strategy | SIR or SIS? who are hubs and bridges? |
| misinformation | one exposure or repeated exposure? directed graph? |
| health campaign | threshold? which groups need reinforcement? |
| Web ranking | which random-walk distribution is meaningful? |
| supply-chain failure | directed dependencies? irreversible failures? |

# Module Takeaway

> Diffusion modeling is model selection. Before computing a metric, decide what exposure means, whether the state sticks, and which topology the process can actually use.

# Wrap-Up

| Question | Modeling choice |
|---|---|
| What spreads? | signal, attention, behavior, infection, failure |
| What changes a node? | flow, one exposure, threshold, infection |
| Does it stick? | SI, SIS, SIR, behavior change, decay |
| Which topology matters? | hubs, bridges, clusters, communities, direction, timing |
| What is the outcome? | score, reach, peak, persistence, final size |

The same graph can accelerate, block, localize, or sustain a process. Network dynamics require matching the spreading rule to the structure.

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